

Project Requirement Document

Software-Based Electrical Fault Prevention & Prediction System

System Architecture Overview

The system follows a modular, layered architecture, ensuring clarity, scalability, and ease of demonstration.

4.1 Electrical Data Simulator (Digital Twin Layer)

This module acts as a virtual electrical system, replacing real-world electrical hardware.

Simulated Parameters :

- Voltage (V)
- Current (A)
- Power (W)
- Temperature (°C)
- Leakage current
- Load imbalance
- Time of usage

Simulation Characteristics

- Normal operating conditions
- Gradual fault development
- Sudden fault spikes
- Day-wise and usage-based variations

The simulator produces continuous data streams that resemble real electrical behavior.

4.2 Fault Injection Engine

This module allows intentional creation of faults in software to test system response.

Supported Fault Types

- Overload conditions
- Loose electrical connections
- Insulation degradation
- Earth leakage
- Early-stage short circuit
- Aging wire effects

This feature makes the system suitable for controlled experiments, demonstrations, and academic evaluation.

4.3 AI / ML Prediction Layer

This layer analyzes simulated and historical data to detect anomalies and predict failures.

Analysis Approaches

- Rule-based logic for basic demonstrations
- Anomaly detection models (e.g., Isolation Forest)
- Time-series prediction models (e.g., LSTM)
- Pattern deviation models (e.g., Autoencoders)

Outputs

- Risk score (0–100%)
- Fault type probability

- Estimated time-to-failure

4.4 Daily Prevention Engine

The Daily Prevention Engine continuously evaluates system health and generates preventive alerts.

Execution Schedule

- Hourly monitoring
- Daily trend analysis
- Pre-peak load evaluation

Sample Alerts

- “Overheating trend detected over the past 3 days”
- “Loose connection likely in Circuit 2”
- “Overload risk predicted within the next 48 hours”

4.5 GenAI Explanation Module

Instead of showing raw technical alerts, this module converts system outputs into human-readable maintenance advice.

Example Output

“The wiring temperature has been increasing abnormally, which usually indicates loose connections. Inspection is recommended within the next two days to avoid fire risk.”

This improves usability for non-technical users.

5. Dashboard & User Interface

A secure web-based dashboard provides real-time visualization and system insights.

Dashboard Features :

- Live simulated electrical readings
- Fault timeline visualization
- Risk meter and alerts
- Historical trend graphs
- Preventive maintenance suggestions

6. Role-Based Access & Authentication

The system implements standard role-based authentication to ensure data privacy and usability.

User Roles

1. Owner (Super Admin)

- Full dashboard access
- Sensitive system information
- User management (add/remove users)

2. Manager

- Overall system health
- Risk summaries and fault alerts
- Historical trends

3. Electrician

- Fault details and locations
- Actionable repair recommendations

4. Floor Engineer

- Floor-wise live status
- Load and temperature trends

5. Site Engineer

- Site-level overview
- Critical warnings and maintenance advice

Authentication Method

- Username and password login
- Password hashing (bcrypt)
- JWT or session-based authentication
- Role stored and validated from database